

Relational Data Languages

Union

- $R \cup S$ is a Binary Operation
- R and S should be type compatible
 - R and S should have same number of attributes
 - Each pair of corresponding attributes must be type compatible (have same or compatible domains)
- Tuples present in R or S or both are retrieved.
- Duplicate tuples are eliminated.

Union Example

- Find details of the books that were published in 2009 or belongs to NEWS category

ISBN	Title	Year	Category	Publ_code
B111	FISH	2007	ARTICLE	P010
B112	GLOW	2009	ARTICLE	P212
B110	FERT	2010	NEWS	P010
B113	FINE ARTS	2009	NEWS	P010
B114	INDU - THE MAID	2008	NOVEL	P201

```
Result1 <- ( $\sigma_{\text{year}=2009}$  (Book))
```

```
Result2 <- ( $\sigma_{\text{Category}=\text{'NEWS'}}$  (Book))
```

```
Result <- Result1 U Result2
```

ISBN	Title	Year	Category	Publ_code
B112	GLOW	2009	ARTICLE	P212
B110	FERT	2010	NEWS	P010
B113	FINE ARTS	2009	NEWS	P010

Consider a query to find the set of all courses taught in the Fall 2009 semester, the Spring 2010 semester, or both.

<i>course_id</i>	<i>sec_id</i>	<i>semester</i>	<i>year</i>	<i>building</i>	<i>room_number</i>	<i>time_slot_id</i>
BIO-101	1	Summer	2009	Painter	514	B
BIO-301	1	Summer	2010	Painter	514	A
CS-101	1	Fall	2009	Packard	101	H
CS-101	1	Spring	2010	Packard	101	F
CS-190	1	Spring	2009	Taylor	3128	E
CS-190	2	Spring	2009	Taylor	3128	A
CS-315	1	Spring	2010	Watson	120	D
CS-319	1	Spring	2010	Watson	100	B
CS-319	2	Spring	2010	Taylor	3128	C
CS-347	1	Fall	2009	Taylor	3128	A
EE-181	1	Spring	2009	Taylor	3128	C
FIN-201	1	Spring	2010	Packard	101	B
HIS-351	1	Spring	2010	Painter	514	C
MU-199	1	Spring	2010	Packard	101	D
PHY-101	1	Fall	2009	Watson	100	A

$$\Pi_{course_id} (\sigma_{semester = \text{“Fall”} \wedge year = 2009} (section)) \cup \Pi_{course_id} (\sigma_{semester = \text{“Spring”} \wedge year = 2010} (section))$$

Figure 6.4 The *section* relation.

Consider a query to find the set of all courses taught in the Fall 2009 semester, the Spring 2010 semester, or both.

course_id	sec_id	semester	year	building	room_number	time_slot_id
BIO-101	1	Summer	2009	Painter	514	B
BIO-301	1	Summer	2010	Painter	514	A
CS-101	1	Fall	2009	Packard	101	H
CS-101	1	Spring	2010	Packard	101	F
CS-190	1	Spring	2009	Taylor	3128	E
CS-190	2	Spring	2009	Taylor	3128	A
CS-315	1	Spring	2010	Watson	120	D
CS-319	1	Spring	2010	Watson	100	B
CS-319	2	Spring	2010	Taylor	3128	C
CS-347	1	Fall	2009	Taylor	3128	A
EE-181	1	Spring	2009	Taylor	3128	C
FIN-201	1	Spring	2010	Packard	101	B
HIS-351	1	Spring	2010	Painter	514	C
MU-199	1	Spring	2010	Packard	101	D
PHY-101	1	Fall	2009	Watson	100	A

$$\Pi_{course_id} (\sigma_{semester = \text{“Fall”} \wedge year = 2009} (section)) \cup \Pi_{course_id} (\sigma_{semester = \text{“Spring”} \wedge year = 2010} (section))$$



course_id
CS-101
CS-315
CS-319
CS-347
FIN-201
HIS-351
MU-199
PHY-101

Figure 6.4 The *section* relation.

Intersection

- Find the books that were published in 2009 and belongs to NEWS category

ISBN	Title	Year	Category	Publ_code
B111	FISH	2007	ARTICLE	P010
B112	GLOW	2009	ARTICLE	P212
B110	FERT	2010	NEWS	P010
B113	FINE ARTS	2009	NEWS	P010
B114	INDU – THE MAID	2008	NOVEL	P201



Result1 <- ($\sigma_{\text{year}=2009}$ (Book))

Result2 <- ($\sigma_{\text{Category}=\text{'NEWS'}}$ (Book))

Result <- Result1 $\cap$ Result2

ISBN	Title	Year	Category	Publ_code
B113	FINE ARTS	2009	NEWS	P010

Set Difference

- To find tuples that are in one relation but are not in another.
- Denoted by -
- $r-s$ produces a relation containing those tuples in r but not in s
- find all the courses taught in the Fall 2009 semester but not in Spring 2010 semester
- As with the union operation, we must ensure that set differences are taken between compatible relations.

$$\Pi_{course_id} (\sigma_{semester="Fall" \wedge year=2009} (section)) - \Pi_{course_id} (\sigma_{semester="Spring" \wedge year=2010} (section))$$



<i>course_id</i>
CS-347
PHY-101

- Find details of the books that were published in 2009 and does not belong to NEWS category

ISBN	Title	Year	Category	Publ_code
B111	FISH	2007	ARTICLE	P010
B112	GLOW	2009	ARTICLE	P212
B110	FERT	2010	NEWS	P010
B113	FINE ARTS	2009	NEWS	P010
B114	INDU – THE MAID	2008	NOVEL	P201

- Find details of the books that were published in 2009 and does not belong to NEWS category
- $\text{Result1} < \sigma_{\text{year}=2009} (\text{Book})$
- $\text{Result2} < \sigma_{\text{Category}=\text{'NEWS'}} (\text{Book})$
- $\text{Result Result1} - \text{Result2}$

ISBN	Title	Year	Category	Publ_code
B111	FISH	2007	ARTICLE	P010
B112	GLOW	2009	ARTICLE	P212
B110	FERT	2010	NEWS	P010
B113	FINE ARTS	2009	NEWS	P010
B114	INDU - THE MAID	2008	NOVEL	P201



ISBN	Title	Year	Category	Publ_code
B112	GLOW	2009	ARTICLE	P212

Properties of Union, Intersect, Difference

- Commutative
 - Satisfied by Union and Intersect
- Associative
 - Satisfied by Union and Intersect
- Distributive
 - $R \cup (S - T) = (R \cup S) - (R \cup T)$

Cartesian Product

The Cartesian-product operation, denoted by a cross ($\times$).

It combine information from any two relations.

Cartesian product of relations r_1 and r_2 is $r_1 \times r_2$.

Cartesian Product

- Combine tuples from two different relations
- Combinatorial manner
- $R \times S$
- $R(A_1, A_2, \dots, A_n) \times S(B_1, B_2, \dots, B_m)$
- $Q(A_1, A_2, \dots, A_n, B_1, B_2, \dots, B_m)$ is the result
- Number of columns in Q
 - $c_Q = c_R + c_S$
- Number of tuples in Q
 - $n_Q = n_R * n_S$

Cartesian Product

ISBN	Title	Category	Publ_code
B111	FISH	ARTICLE	P010
B112	GLOW	ARTICLE	P212
B110	FERT	NEWS	P010

Publ_code	Name	Address
P011	Pub1	Add1
P212	Pub2	Add2
P010	Pub3	Add3

Book

X

Publisher

ISBN	Title	Category	Publ_code
B111	FISH	ARTICLE	P010
B111	FISH	ARTICLE	P010
B111	FISH	ARTICLE	P010
B112	GLOW	ARTICLE	P212
B112	GLOW	ARTICLE	P212
B112	GLOW	ARTICLE	P212

Publ_code	Name	Address
P011	Pub1	Add1
P212	Pub2	Add2
P010	Pub3	Add3
P011	Pub1	Add1
P212	Pub2	Add2
P010	Pub3	Add3

Example.

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

The Instructor relation

<i>ID</i>	<i>course_id</i>	<i>sec_id</i>	<i>semester</i>	<i>year</i>
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

The teaches relation

instructor X teaches

relation schema for r as:

(instructor.ID, name, dept_name, salary
teaches.ID, course_id, sec_id, semester, year)

[illegible]

Find the names of all instructors in the Physics department together with the course id of all courses they taught

$$\sigma_{dept_name = \text{"Physics"}}(instructor \times teaches)$$
[illegible]

$\sigma_{instructor.ID = teaches.ID}(\sigma_{dept_name = \text{"Physics"}}(instructor \times teaches))$

$\Pi_{name, course_id}(\sigma_{instructor.ID = teaches.ID}(\sigma_{dept_name = \text{"Physics"}}(instructor \times teaches)))$

<i>name</i>	<i>course_id</i>
Einstein	PHY-101

$\Pi_{name, course_id} (\sigma_{instructor.ID = teaches.ID} (\sigma_{dept_name = \text{“Physics”}} (instructor \times teaches)))$.

OR

$\Pi_{name, course_id} (\sigma_{instructor.ID = teaches.ID} ((\sigma_{dept_name = \text{“Physics”}} (instructor)) \times teaches))$